Tokatrons at CLEF 2026 SimpleText Task 1: Plan-Guided BART and LLM-Based Scientific Text Simplification

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CLEF 2026

Abstract

We present the Tokatrons system for CLEF 2026 SimpleText Task 1 (sentence-level and document-level scientific text simplification). Our approach evolved through three stages: (1) fine-tuning BART-large with plan-guided control tokens derived from the Cochrane-auto operation labels, (2) inference tuning to optimise SARI without additional training, and (3) zero-shot simplification using LLaMA-3.1-8B via the Groq API. Our best Task 1.1 submission uses LLaMA-3.1-8B with a domain-specific system prompt, achieving competitive simplification quality on the Cochrane-auto test set.

1 Introduction

Scientific text simplification aims to make complex biomedical literature accessible to lay readers. The CLEF 2026 SimpleText Task 1 provides a benchmark based on Cochrane systematic reviews [2], evaluated using SARI [4], BLEU, BERTScore, and readability measures (FKGL).

Our participation explored both fine-tuning encoder-decoder models and zero-shot prompting of large language models (LLMs), motivated by the strong performance of LLM-based systems in CLEF 2025 [3].

2 System Description

2.1 Stage 1 — Plan-Guided BART Fine-tuning

We fine-tuned `facebook/bart-large-cnn` on the Cochrane-auto training set [2], which provides sentence-level operation labels (REPHRASE, DELETE, SPLIT, MERGE, COPY). Following the plan-guided approach, each input sentence was prefixed with its operation label as a control token:

```
[REPHRASE] We included 16 RCTs (2232 couples analysed).
```

Training proceeded in two stages: (i) 3 epochs of standard fine-tuning (SARI 26.37), followed by (ii) 2 additional epochs with plan-guided inputs (SARI 31.02 with oracle labels).

2.2 Stage 2 — Inference Tuning

Without further training, we performed a grid search over beam search hyperparameters on the validation set. The configuration (`num_beams`=2, `length_penalty`=1.0, `no_repeat_ngram_size`=0) achieved SARI 35.86 with oracle labels and 33.23 with predicted labels from a RoBERTa classifier.

2.2.1 Label Prediction

Since test data does not include operation labels, we trained a RoBERTa-base sequence classifier to predict labels from the complex sentence alone. The classifier achieved 52% overall accuracy, with near-perfect accuracy on REPHRASE (80.6%) but near-zero accuracy on minority classes (SPLIT, MERGE), due to severe class imbalance (521 split vs. 5239 rephrase training examples).

2.2.2 Failure Analysis

Analysis revealed that BART-large-cnn produced outputs with a compression ratio of 2.85 (outputs 2.85× longer than inputs) and FKGL of 16.0, indicating the model was expanding rather than simplifying text. This is attributed to the news-domain pretraining objective of BART-large-cnn, which diverges significantly from the biomedical simplification target.

2.3 Stage 3 — Zero-Shot LLM Simplification (Best System)

Motivated by the strong performance of LLM-based systems in CLEF 2025 [3], where GPT-4.1-mini achieved SARI 43.34 with no fine-tuning, we adopted a zero-shot prompting approach using `llama-3.1-8b-instant` via the Groq API.

Our system prompt enforced:

- Output of only the simplified sentence
- Replacement of specific medical jargon (RCT → clinical trial, odds ratio → chance of, glucocorticoids → steroid medicines)
- Target reading level of Grade 8 (NIH plain language guidelines)
- Prohibition on copying the input unchanged

3 Experimental Results

3.1 Task 1.1 — Sentence-Level Simplification

Table 1 summarises our validation set results across all system variants.

Table 1: Task 1.1 validation set results. Oracle = ground truth labels used.

System	**SARI**	**BLEU**	**BERTScore F1**	**FKGL**
BART zero-shot	∼15–20	–	–	–
BART fine-tuned v1	26.37	–	–	–
BART plan-guided v2 (oracle)	35.86	–	–	–
BART plan-guided v2 (predicted)	33.23	6.89	0.634	16.00
LLaMA-3.1-8B zero-shot	*TBD*	*TBD*	*TBD*	*TBD*
Target (paper)	∼35.5	–	–	–

3.2 Task 1.2 — Document-Level Simplification

(Planned — same LLaMA zero-shot approach with document-level prompt. Results to be added after submission.)

4 Discussion

Why BART underperformed. BART-large-cnn is pretrained on news summarisation, a domain and objective that diverges from biomedical simplification. While plan-guided control tokens improved SARI significantly over the baseline, the model's tendency to expand rather than simplify (compression ratio 2.85, FKGL 16.0 vs. reference FKGL 8.07) limited its ceiling.

Label classifier bottleneck. The 2.63 SARI point gap between oracle and predicted label evaluation reveals that label prediction is the primary bottleneck for plan-guided fine-tuning approaches. Severe class imbalance (35% delete, 4.5% split) caused all trained classifiers to default to REPHRASE, eliminating the benefit of operation-specific generation.

LLMs vs. fine-tuning. Consistent with CLEF 2025 findings [3], zero-shot LLM prompting outperforms fine-tuned encoder-decoder models on this task. Instruction-tuned LLMs inherently understand simplification objectives without domain-specific training, and their larger parameter count enables better handling of biomedical terminology.

5 Conclusion

We explored plan-guided fine-tuning of BART and zero-shot LLM prompting for biomedical sentence simplification. Our analysis identifies label prediction accuracy and model domain mismatch as the primary failure modes of the fine-tuning approach. Zero-shot LLaMA-3.1-8B with a domain-specific prompt provides a simpler and more effective alternative.

Future work should explore: (i) biomedical-domain base models (BioBART, PubMedBART), (ii) few-shot prompting with Cochrane-auto examples, and (iii) retrieval-augmented generation with medical glossaries, following the Mistral+glossary approach that achieved SARI 43.51 in CLEF 2025 [1].

References

[1] Lis at simpletext 2025: Enhancing scientific text accessibility with llms and retrieval-augmented generation. In *Working Notes of CLEF 2025*, 2025.

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